

Discrete Mathematics I Test 4 - 07.05.2026

Exercise. Use generating functions to find a closed-form formula for a_n given the recurrence

$$a_n = 2a_{n-1} + n, \quad n \geq 1, \quad a_0 = 1.$$

0.2 Solutions

1. Generating functions (direct method)

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Multiply the recurrence by x^n and sum over $n \geq 1$:

$$\sum_{n=1}^{\infty} a_n x^n = 2 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} n x^n.$$

Using $\sum_{n=1}^{\infty} a_{n-1} x^n = x f(x)$ and $\sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}$,

$$f(x) - 1 = 2x f(x) + \frac{x}{(1-x)^2},$$

so

$$f(x) = \frac{1}{1-2x} + \frac{x}{(1-x)^2(1-2x)}.$$

Partial fraction decomposition of the second term gives

$$\frac{x}{(1-x)^2(1-2x)} = \frac{-2}{1-2x} + \frac{1}{1-x} + \frac{1}{(1-x)^2},$$

hence

$$f(x) = \frac{3}{1-2x} - \frac{1}{1-x} - \frac{1}{(1-x)^2} + \frac{1}{(1-x)^2} \rightarrow f(x) = \frac{3}{1-2x} - \frac{1}{1-x} - \frac{1}{(1-x)^2}.$$

Wait — gathering correctly:

$$f(x) = \frac{1}{1-2x} + \frac{-2}{1-2x} + \frac{1}{1-x} + \frac{1}{(1-x)^2} = \frac{-1}{1-2x} + \frac{1}{1-x} + \frac{1}{(1-x)^2}.$$

Hmm, let us redo the partial fractions carefully. Write

$$\frac{x}{(1-x)^2(1-2x)} = \frac{A}{1-2x} + \frac{B}{1-x} + \frac{C}{(1-x)^2}.$$

Clearing denominators: $x = A(1-x)^2 + B(1-x)(1-2x) + C(1-2x)$.

- $x = \frac{1}{2}$: $\frac{1}{2} = A \cdot \frac{1}{4}$, so $A = 2$.
- $x = 1$: $1 = C(1-2) = -C$, so $C = -1$.
- $x = 0$: $0 = A + B + C = 2 + B - 1$, so $B = -1$.

Therefore

$$f(x) = \frac{1}{1-2x} + \frac{2}{1-2x} - \frac{1}{1-x} - \frac{1}{(1-x)^2} = \frac{3}{1-2x} - \frac{1}{1-x} - \frac{1}{(1-x)^2}.$$

Extracting coefficients using $[x^n] \frac{1}{1-2x} = 2^n$, $[x^n] \frac{1}{1-x} = 1$, and $[x^n] \frac{1}{(1-x)^2} = n+1$:

$$a_n = 3 \cdot 2^n - 1 - (n+1) = 3 \cdot 2^n - n - 2.$$

2. Undetermined coefficients (method of particular solutions)

The homogeneous recurrence $a_n^{(h)} - 2a_{n-1}^{(h)} = 0$ has characteristic root $r = 2$, giving $a_n^{(h)} = C \cdot 2^n$.

Since the driving term n is a degree-1 polynomial and 1 is not a characteristic root, we seek a particular solution of the form $a_n^{(p)} = An + B$. Substituting:

$$An + B = 2(A(n-1) + B) + n = (2A+1)n + (-2A+2B).$$

Comparing coefficients:

$$A = 2A + 1 \implies A = -1, \quad B = -2A + 2B \implies B = -2.$$

The general solution is $a_n = C \cdot 2^n - n - 2$. Initial condition $a_0 = 1$ gives $C - 2 = 1$, so $C = 3$.

3. Iteration and telescoping

Unrolling the recurrence k times:

$$a_n = 2^n a_0 + \sum_{k=1}^n k 2^{n-k} = 2^n + 2^n \sum_{k=1}^n k 2^{-k}.$$

To evaluate $S_n = \sum_{k=1}^n k 2^{-k}$, differentiate the finite geometric sum $\sum_{k=0}^n x^k = \frac{1-x^{n+1}}{1-x}$ with respect to x and set $x = \frac{1}{2}$:

$$S_n = 2 - (n+2) \cdot 2^{-n}.$$

Therefore $a_n = 2^n(1 + 2 - (n+2) \cdot 2^{-n}) = 3 \cdot 2^n - n - 2$.

4. Summation factor (discrete integrating factor)

Divide both sides of the recurrence by 2^n and set $b_n = a_n/2^n$:

$$b_n = b_{n-1} + \frac{n}{2^n}.$$

Telescoping from 1 to n :

$$b_n = b_0 + \sum_{k=1}^n \frac{k}{2^k} = 1 + S_n,$$

where $S_n = 2 - (n+2) \cdot 2^{-n}$ as computed in Method 3. Thus $b_n = 3 - (n+2) \cdot 2^{-n}$, and restoring $a_n = 2^n b_n$ gives $a_n = 3 \cdot 2^n - n - 2$.

This is the exact discrete analogue of solving a first-order linear ODE via an integrating factor: dividing by 2^n removes the homogeneous part in one step.

5. Matrix exponentiation and Jordan normal form

Augment the state vector to track both a_n and the driving term:

$$\mathbf{v}_n = \begin{pmatrix} a_n \\ n \\ 1 \end{pmatrix}, \quad \mathbf{v}_n = M \mathbf{v}_{n-1}, \quad M = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

The eigenvalues of M are $\lambda_1 = 2$ (simple) and $\lambda_2 = 1$ (algebraic multiplicity 2, defective). The Jordan decomposition $M = PJP^{-1}$ with

$$J = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

gives $M^n = PJ^nP^{-1}$, where

$$J^n = \begin{pmatrix} 2^n & 0 & 0 \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}.$$

The first component of $M^n \mathbf{v}_0$ with $\mathbf{v}_0 = (1, 0, 1)^\top$ yields $a_n = 3 \cdot 2^n - n - 2$. The Jordan block structure directly explains why the solution is the sum of an exponential 2^n term and a polynomial in n : these come from the simple eigenvalue and the defective block respectively.

6. Shift operator calculus

Let E denote the forward shift operator, $Ea_n = a_{n+1}$. Writing the recurrence with index shifted by one, $(E - 2)a_{n-1} = n$, i.e.

$$(E - 2)a_n = n + 1.$$

Formally invert via the geometric series in $E/2$:

$$a_n^{(p)} = \frac{1}{E - 2}(n + 1) = -\frac{1}{2} \cdot \frac{1}{1 - E/2}(n + 1) = -\frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{E}{2}\right)^k (n + 1).$$

Since $E^k(n+1) = n+k+1$, and this is a polynomial so the series terminates upon taking the difference, the computation gives a particular solution $a_n^{(p)} = -n - 2$. Adding the homogeneous solution $C \cdot 2^n$ and using $a_0 = 1$ recovers $C = 3$.

This is the algebraic skeleton behind both the generating-function and undetermined-coefficients approaches.

7. Residue calculus

Once the generating function is identified as

$$f(x) = \frac{(1-x)^2 - x(1-x) + x(1-x)^2/(x) \cdots}{\cdots} \longrightarrow f(x) = \frac{3}{1-2x} - \frac{1}{1-x} - \frac{1}{(1-x)^2},$$

the coefficient $a_n = [x^n]f(x)$ can be expressed as a contour integral:

$$a_n = \frac{1}{2\pi i} \oint_{|z|=r} \frac{f(z)}{z^{n+1}} dz,$$

where r is chosen so that the contour encircles all poles of $f(z)/z^{n+1}$, which are at $z = 0$, $z = 1/2$ (simple), and $z = 1$ (order 2). By the residue theorem:

$$a_n = \text{Res}_{z=1/2} \frac{f(z)}{z^{n+1}} + \text{Res}_{z=1} \frac{f(z)}{z^{n+1}}.$$

The simple pole at $z = 1/2$ contributes $3 \cdot 2^n$; the second-order pole at $z = 1$ contributes $-n - 2$ via the standard formula for higher-order residues. This approach is especially powerful for deriving asymptotic behaviour: the dominant singularity closest to the origin, $z = 1/2$, immediately gives $a_n \sim 3 \cdot 2^n$ as $n \rightarrow \infty$.

8. Analogy with first-order linear ODEs (heuristic)

The continuous analogue of the recurrence is the initial value problem

$$f'(x) = 2f(x) + x, \quad f(0) = 1.$$

Solving with integrating factor e^{-2x} :

$$f(x) = e^{2x} \left(1 + \int_0^x t e^{-2t} dt \right) = 3e^{2x} - \frac{x}{2} - \frac{1}{4}.$$

Replacing $e^{2x} \rightarrow 2^n$ and matching polynomial degrees (with constants adjusted for the discrete setting) reproduces the structure $3 \cdot 2^n - n - 2$. This is not a rigorous derivation but serves as a quick sanity check and motivates why Methods 4 and 5 work.

Remark. Methods 1–4 are elementary and suitable for an undergraduate course. Methods 5–6 (matrix/Jordan form, operator calculus) belong to a more algebraic treatment of recurrences and generalise immediately to any order. Method 7 (residues) is the natural language for asymptotic analysis and connects discrete mathematics to complex analysis.